

Mathematical Analysis II

Week 4 Homework (March 26)

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Section 9.2

19. 求下列复合函数的偏导数或导数.

(2) 设 $u = e^{xyz}$, $x = rs$, $y = \frac{r}{s}$, $z = r^s$, 求 $\frac{\partial u}{\partial r}$, $\frac{\partial u}{\partial s}$;

解. 将 $x = rs$, $y = \frac{r}{s}$, $z = r^s$ 代入 $u = e^{xyz}$, 得

$$u = e^{r^{s+2}}.$$

因此

$$\begin{aligned}\frac{\partial u}{\partial r} &= (s+2)r^{s+1}e^{r^{s+2}}, \\ \frac{\partial u}{\partial s} &= r^{s+2}e^{r^{s+2}} \ln r.\end{aligned}$$

(4) 设 $u = \frac{e^{ax}(y-z)}{a^2+1}$, $y = a \sin x$, $z = \cos x$, 求 $\frac{du}{dx}$;

解. 将 $y = a \sin x$, $z = \cos x$ 代入 $u = \frac{e^{ax}(y-z)}{a^2+1}$, 得

$$u = \frac{e^{ax}(a \sin x - \cos x)}{a^2 + 1}.$$

因此

$$\frac{du}{dx} = \frac{ae^{ax}(a \sin x - \cos x) + e^{ax}(a \cos x + \sin x)}{a^2 + 1} = e^{ax} \sin x.$$

20. 求下列复合函数的偏导数或导数, 其中各题中的 f 均有连续的二阶偏导.

(2) 设 $u = f(x, y, z)$, $x = \sin t$, $y = \cos t$, $z = e^t$, 求 $\frac{du}{dt}$;

解. 由链式法则得

$$\frac{du}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt} = f_x \cos t - f_y \sin t + f_z e^t.$$

(4) 设 $u = f(x + y + z, x^2 + y^2 + z^2)$, 求 $\frac{\partial u}{\partial x}$, $\frac{\partial^2 u}{\partial x^2}$, $\frac{\partial^2 u}{\partial x \partial y}$;

解. 令 $f = f(s, t)$, 其中 $s = x + y + z, t = x^2 + y^2 + z^2$.

首先计算一阶偏导数:

$$\frac{\partial u}{\partial x} = \frac{\partial f}{\partial s} \frac{\partial s}{\partial x} + \frac{\partial f}{\partial t} \frac{\partial t}{\partial x} = f_s + 2xf_t.$$

再计算二阶偏导数:

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \right) = \frac{\partial f_s}{\partial x} + 2f_t + 2x \frac{\partial f_t}{\partial x} = f_{ss} + 2xf_{st} + 2f_t + 2xf_{ts} + 4x^2 f_{tt} = f_{ss} + 4xf_{st} + 2f_t + 4x^2 f_{tt}.$$

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial f_s}{\partial y} + 2x \frac{\partial f_t}{\partial y} = f_{sy} + 2yf_{st} + 2xf_{ty} + 4xyf_{tt} = f_{ss} + 2(x+y)f_{st} + 4xyf_{tt}.$$

22. 试求函数 $z = \arctan \frac{y}{x}$ 在圆 $x^2 + y^2 - 2x = 0$ 上一点 $P\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$ 处沿该圆周逆时针方向上的方向微商.

解. 圆的标准方程为

$$(x-1)^2 + y^2 = 1.$$

因此当 $t = \frac{2\pi}{3}$ 时, 对应点 $P\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$.

首先计算偏导数:

$$\frac{\partial z}{\partial x} = -\frac{y}{x^2 + y^2}, \quad \frac{\partial z}{\partial y} = \frac{x}{x^2 + y^2}.$$

在点 P 处, $x^2 + y^2 = 1$, 因此梯度为

$$\nabla z|_P = \left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right).$$

圆的法向量为 $\vec{n} = (x-1, y)$, 因此逆时针切向量为

$$\vec{\tau} = (-y, x-1) = \left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right).$$

因此方向微商为

$$\frac{\partial z}{\partial \tau} = \nabla z \cdot \vec{\tau} = \frac{3}{4} - \frac{1}{4} = \frac{1}{2}.$$

23. 求函数 $u = x^2 + 2y^2 + 3z^2 + xy + 3x - 2y - 6z$ 在点 $(1, 1, -1)$ 的梯度和最大方向微商.

解. 首先计算一阶偏导数:

$$\frac{\partial u}{\partial x} = 2x + y + 3, \quad \frac{\partial u}{\partial y} = 4y + x - 2, \quad \frac{\partial u}{\partial z} = 6z - 6.$$

在点 $(1, 1, -1)$ 处计算得:

$$\left. \frac{\partial u}{\partial x} \right|_{(1,1,-1)} = 6, \quad \left. \frac{\partial u}{\partial y} \right|_{(1,1,-1)} = 3, \quad \left. \frac{\partial u}{\partial z} \right|_{(1,1,-1)} = -12.$$

因此梯度为

$$\nabla u|_{(1,1,-1)} = (6, 3, -12).$$

最大方向微商为

$$\sqrt{6^2 + 3^2 + 12^2} = \sqrt{189}.$$

26. 设 $z = f(xy)$, f 为可微函数. 证明 $x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y} = 0$.

证. 计算偏导数得

$$\frac{\partial z}{\partial x} = y f'(xy), \quad \frac{\partial z}{\partial y} = x f'(xy).$$

因此

$$x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y} = xy f'(xy) - xy f'(xy) = 0.$$

Q.E.D.

28. 证明函数 $u = \varphi(x - at) + \psi(x + at)$ 满足波动方程

$$\frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2}.$$

其中 φ, ψ 有连续的二阶微商.

证. 首先计算关于 t 的偏导数:

$$\frac{\partial u}{\partial t} = -a\varphi'(x - at) + a\psi'(x + at),$$

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial}{\partial t} \left(\frac{\partial u}{\partial t} \right) = a^2 \varphi''(x - at) + a^2 \psi''(x + at).$$

再计算关于 x 的偏导数:

$$\frac{\partial u}{\partial x} = \varphi'(x - at) + \psi'(x + at),$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \right) = \varphi''(x - at) + \psi''(x + at).$$

因此得到

$$\frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2}.$$

Q.E.D.

第 9 章综合习题

1. 设 $a_1, a_2, \dots, a_n$ 是非零常数. $f(x_1, x_2, \dots, x_n)$ 在 $\mathbb{R}^n$ 上可微. 求证: 存在 $\mathbb{R}$ 上一元可微函数 $F(s)$ 使得 $f(x_1, x_2, \dots, x_n) = F(a_1 x_1 + a_2 x_2 + \dots + a_n x_n)$ 的充分必要条件是

$$a_j \frac{\partial f}{\partial x_i} = a_i \frac{\partial f}{\partial x_j}, \quad i, j = 1, 2, \dots, n.$$

证. 令 $s = a_1 x_1 + a_2 x_2 + \dots + a_n x_n$

• 必要性

因为存在 $\mathbb{R}$ 上一元可微函数 $F(s)$ 使得 $f(x_1, x_2, \dots, x_n) = F(a_1 x_1 + a_2 x_2 + \dots + a_n x_n)$, 对二者求微分, 得 $\frac{\partial f}{\partial x_i} = a_i F'(s)$

所以 $a_j \frac{\partial f}{\partial x_i} = a_j a_i F'(s) = a_i a_j F'(s) = a_i \frac{\partial f}{\partial x_j}$

- 充分性

因为 $a_j \frac{\partial f}{\partial x_i} = a_i \frac{\partial f}{\partial x_j}$, $i, j = 1, 2, \dots, n$, 可以得到 $\frac{1}{a_i} \frac{\partial f}{\partial x_i}$ 与 i 无关

令 $\varphi(x_1, \dots, x_n) = \frac{1}{a_i} \frac{\partial f}{\partial x_i}$

所以 $\frac{\partial f}{\partial x_i} = a_i \varphi$

所以 $df = \sum_{i=1}^n a_i \varphi dx_i = \varphi ds$